

Zixi Chen

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Research Interest

Underactuated dexterous hand; Surgical robotics; Continuum robotics; Vision-based tactile sensing

Research and Working Experience

Vrije Universiteit Brussel, Belgium

Sep. 2026 – Present

Postdoctoral Researcher

Collaborating Professors: Bram Vanderborght

Responsibilities: Assistance in coordinating EU project RESSORT (€9.7 M); Research on underactuated dexterous hand

Ultimage Intelligent Technology (Macao) Ltd., Macao (CN)

Oct. 2025 – Present

Scientific Advisor

Collaborating Professor: Jia Gu

Responsibilities: Development of continuum surgical robotics; Research on rolling joint continuum robotics and surgical robot teleoperator; PhD student supervision

King's College London and Tsinghua University, China

Nov. 2021 – Sep. 2022

Research Assistant

Collaborating Professor: Shan Luo, Bin Fang

Responsibilities: Physics-based optical tactile sensor simulation; RL-based plasticine manipulation

Education

Scuola Superiore Sant'Anna, Italy

Oct. 2022 – Oct. 2025

Ph.D. in Biorobotics (with distinction) **Supervisor:** Cesare Stefanini, Arianna Menciassi (**IEEE Fellow**)

Thesis: Data-driven Control Approaches for Modular Soft Robots

EPFL, Switzerland

Aug. 2024 – Oct. 2024

Visiting Ph.D. student

Supervisor: Josie Hughes

Imperial College London, United Kingdom

Oct. 2020 – Oct. 2021

M.Sc. in Control Systems (with distinction)

Supervisor: Tae-kyun Kim

Thesis: Data augmentation by adversarial learning

Harbin Institute of Technology (Shenzhen), China

Aug. 2016 – Jun. 2020

B.Eng. in Mechanical Engineering (Outstanding Graduate)

Supervisor: Qingbin Gao

Thesis: Gait Control and Stability Analysis of Biped Robot Walking on a Slope

Publications and Patents

Citations: 664 | h-index: 10 (Google Scholar, Sep 2026)

Five Selected Publications:

- [1] **Z. Chen**, Q. Guan, J. Hughes, et al., "A Versatile Neural Network Configuration Space Planning and Control Strategy for Modular Soft Robot Arms," in **IEEE Transactions on Robotics**, vol. 41, pp. 4269-4282, 2025.
- [2] **Z. Chen**, X. Ren, Y. Hamamatsu, et al., "AdapJ: An Adaptive Extended Jacobian Controller for Soft Manipulators," in **IEEE/ASME Transactions on Mechatronics**, vol. 31, no. 2, pp. 1753-1763, 2026.

- [3] **Z. Chen**, F. Renda, A. Le Gall, et al., "Data-driven Methods Applied to Soft Robot Modeling and Control: A Review," in *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 2241-2256, 2025.
- [4] **Z. Chen**, D. Wu, Q. Guan, D. Hardman, F. Renda, J. Hughes, T. G. Thuruthe, C. Della Santina, B. Mazzolai, H. Zhao, and C. Stefanini, "A Survey on Soft Robot Adaptability: Implementations, Applications, and Prospects," in *IEEE Robotics & Automation Magazine*, vol. 33, no. 2, pp. 183-195, 2026.
- [5] **Z. Chen**, M. Bernabei, V. Mainardi, et al., "A Novel and Accurate BiLSTM Configuration Controller for Modular Soft Robots with Module Number Adaptability," in *Soft Robotics* (accepted)

Other Publications:

- [6] Y. Hamamatsu, **Z. Chen**, M. Kruusmaa, et al., "Strouhal-Aware Model Predictive Control for Efficient Multi-Fin Flapping Locomotion" in *IEEE Journal of Oceanic Engineering* (accepted)
- [7] Y. Shao, Y. Zhang, J. Gu, **Z. Chen**, et al., "Deep-Learning-Based Control of a Decoupled Two-Segment Continuum Robot for Endoscopic Submucosal Dissection" in *IEEE/ASME Transactions on Mechatronics* (accepted)
- [8] Y. Sun, S. Zhang, **Z. Chen**, et al., "Soft Contact Simulation and Manipulation Learning of Deformable Objects with Vision-based Tactile Sensor," in *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 17618-17630, 2025.
- [9] S. Jing, T. Yao, K. Zhang, D. Wu, Q. Wang, **Z. Chen**, et al., "Ultrasound-Guided Robotic Blood Drawing and In Vivo Studies on Submillimetre Vessels of Rats," in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10481-10486, 2025.
- [10] Z. Shen, Y. Sun, S. Zhang, **Z. Chen**, et al., "Simulation of Optical Tactile Sensors Supporting Slip and Rotation Using Path Tracing and IMPM," in *IEEE Robotics and Automation Letters*, vol. 9, no. 12, pp. 11218-11225, 2024.
- [11] W. Liu, Y. Shao, Y. Zhang, **Z. Chen**, et al., "Desectbot: Design and validation of a novel two-segment decoupled continuum robotic system for endoscopic submucosal dissection," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 6983-6989, 2024.
- [12] A. Le Gall, T. Dangel, Z. Chen, et al., "An origami-based fluidic actuator for Minimally Invasive Surgery," in *International Conference and Exhibition on New Actuator Systems and Applications*, pp. 260-262, 2024.
- [13] **Z. Chen**, X. Ren, M. Bernabei, et al., "A Hybrid Adaptive Controller for Soft Robot Interchangeability," in *IEEE Robotics and Automation Letters*, vol. 9, no. 1, pp. 875-882, 2024.
- [14] S. Zhang, Y. Sun, J. Shan, **Z. Chen**, et al., "TIRgel: A Visuo-Tactile Sensor with Total Internal Reflection Mechanism for External Observation and Contact Detection," in *IEEE Robotics and Automation Letters*, vol. 8, no. 10, pp. 6307–6314, 2023.
- [15] **Z. Chen**, S. Zhang, S. Luo, et al., "Tacchi: A Pluggable and Low Computational Cost Elastomer Deformation Simulator for Optical Tactile Sensors," in *IEEE Robotics and Automation Letters*, vol. 8, no. 3, pp. 1239–1246, 2023.
- [16] S. Zhang*, **Z. Chen***, Y. Gao, et al., "Hardware Technology of Vision-Based Tactile Sensor: A Review," in *IEEE Sensors Journal*, vol. 22, no. 22, pp. 21410-21427, 2022.

Patents:

- [1] L. Yi, **Z. Chen**, Y. Shao, et al., "Master Manipulator for Intuitive Operation of an Isomorphic Master-Slave Surgical Robotic System," China Utility Model Patent, Application No. 202621315243.9, filed Aug. 24, 2026. (pending)

- [2] L. Yi, **Z. Chen**, Y. Shao, et al., “Anti-Tremor Filtering and Safety-Constrained Control Method for an Isomorphic Master–Slave Natural-Orifice Surgical Robot,” China Invention Patent, Application No. 202611288055.6, filed Aug. 24, 2026. (pending)
- [3] Y. Shao, L. Yi, **Z. Chen**, et al., “Surgical Robot Master Manipulator with Isomorphic Mapping to the Slave Side and Its Control Method,” China Invention Patent, Application No. 202611288059.4, filed Aug. 24, 2026. (pending)

Supervision and Teaching

- Lingxiao Yi (2026), PhD student at City University of Macau. Supervision for planned continuum surgical robotics publication
- Yuhao Sun (2023-2025), master student at Anhui University of Technology. Supervision for TASE paper publication [8].
- Zirong Shen (2024), undergraduate student at Tsinghua University. Supervision for RAL publication [10].

Academic Service

Peer Review Contributions:

npj Robotics, T-RO, TMECH, T-ASE, SoRo, TIE, T-MRB, RAS, RA-L, IROS, ICRA

Workshops:

I am the lead organizer of the serial workshops:

- IROS 2026 workshop 'The SOFT frontier 4: Intelligent Interactions in Soft Robotics.'
- ICRA 2026 workshop 'The SOFT frontier 3: Soft or Rigid? New Trends in Wearable Robotics.'
- IROS 2025 workshop 'The SOFT frontier 2: practical applications in soft robotics.'
- IROS 2024 workshop 'The SOFT frontier: adaptive technologies in soft robotics.'

Awards

National Award for Outstanding Chinese Students Studying Abroad (国家优秀自费留学生奖学金), \$6k, 2025